

Dimensional Reduction in Supergravity and Duality Symmetries

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Overview

- 1 Introduction
- 2 Kaluza-klein reduction of the Einstein-Hilbert action
- 3 Reduction of the NS-NS action
- 4 $O(E, E)$ symmetry
- 5 Conclusion

The objectif

- How does a 4 dimensional observer perceive an universe in $4 + E$ dimension.
- Finding equivalent theory.

justification

Why are we studying that?

Hilbert–Einstein action

We are going to follow Scherk and Schwarz, “How to get masses from extra dimensions” paper for this part.

The Action

$$\begin{aligned}
 S &= \int dx^D \int \frac{dy^E}{\rho(E)} \sqrt{-\hat{g}} \hat{R} \\
 &= \int dx^D \hat{V} \{ \hat{\omega}^{\hat{s}\hat{t}}_{\hat{s}} \hat{\omega}^{\hat{r}}_{\hat{r}\hat{t}} + \hat{\omega}^{\hat{s}\hat{t}}_{\hat{r}} \hat{\omega}^{\hat{r}}_{\hat{s}\hat{t}} + 2 \hat{\omega}^{\hat{r}\hat{s}}_{\hat{s}} \hat{V}^{\hat{\mu}}_{\hat{r}} \partial_{\hat{\mu}} \hat{\phi} \}
 \end{aligned}$$

hat $\rightarrow$ high dimension, Beginning of an alphabet $\rightarrow$ internal space,
end of an alphabet $\rightarrow$ space-time.

Vielbein ansatz

definition

$$\hat{\omega}_{trs} = -\hat{\Omega}_{\hat{t}\hat{r}\hat{s}} + \hat{\Omega}_{\hat{r}\hat{s}\hat{t}} - \hat{\Omega}_{\hat{s}\hat{t}\hat{r}}$$

$$\hat{\Omega}_{\hat{r}\hat{s}\hat{t}} = \frac{1}{2}(\hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{s}}^{\hat{\nu}} - \hat{V}_{\hat{r}}^{\hat{\nu}} \hat{V}_{\hat{s}}^{\hat{\mu}}) \partial_{\hat{\nu}} \hat{V}_{\hat{\mu}\hat{t}}$$

Vielbein ansatz

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} V_{\mu}^r & A_{\mu}^{(1)\beta} \Phi_{\beta}^a \\ 0 & \Phi_{\alpha}^a \end{pmatrix}.$$

Result

Reduced action

$$S = \int dx^D V \delta \left(\frac{-1}{4} g^{\rho\lambda} \partial_\lambda h_{\alpha\beta} \partial_\rho h^{\alpha\beta} + \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^\beta h_{\alpha\beta} + \right. \\ \left. + R + \frac{1}{4} \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \log(h) \right).$$

What do I spy with my little eyes?

NS-NS action

For the second goal, we are going to follow Maharana and Schwarz, “Noncompact symmetries in string theory”.

The action

$$S = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} \left(\hat{R} + \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\mu}} \hat{\phi} - \frac{1}{12} H_{\hat{\mu}\hat{\nu}\hat{\rho}} H^{\hat{\mu}\hat{\nu}\hat{\rho}} \right),$$

Finalization

Reduced action

$$\begin{aligned}
 S = & \int dx^D V e^{-\phi} \left[(R + \partial_\mu \phi \partial^\mu \phi) \right. \\
 & + \left(\frac{1}{4} g^{\mu\nu} (\partial_\mu h_{\alpha\beta} \partial_\nu h^{\alpha\beta} - h^{\alpha\beta} h^{\gamma\delta} \partial_\mu B_{\alpha\gamma} \partial_\nu B_{\beta\delta}) \right) \\
 & + \left(-\frac{1}{4} g^{\mu\rho} g^{\nu\lambda} (h_{\alpha\beta} F_{\mu\nu}^{(1)\alpha} F_{\rho\lambda}^{(1)\beta} + h^{\alpha\beta} H_{\mu\nu\alpha} H_{\rho\lambda\beta}) \right) \\
 & \left. + \left(-\frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right) \right]
 \end{aligned}$$

$O(E, E)$ group

Definition

A matrix M is in $O(E, E)$ if it is a square $E \times E$ matrix and, for the matrix

$$\zeta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which can be used as a metric for the group $O(E, E)$, as it has E negative eigenvalues and E positive eigenvalues in a basis rotated from the one with diagonal metric, the matrix M satisfies

$$M^T \zeta M = \zeta.$$

$O(E, E)$ group action

$$\begin{aligned} A^i &\rightarrow \Omega_j^i A^j, \\ M_{ij} &\rightarrow \Omega_i^l M_{lk} \Omega_j^k, \end{aligned}$$

Enter M

$$M = \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix}$$

An action reformed

The new action

$$\begin{aligned}
 S = \int dx^D V e^{-\phi} & \left[R + g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \right. \\
 & + \frac{1}{8} \text{tr}(\partial_\mu M^{-1} \partial^\mu M) - \frac{1}{4} \mathcal{F}_{\mu\nu}^i (M^{-1})_{ij} \mathcal{F}^{\mu\nu j} \\
 & - \frac{1}{12} \left(\partial_\mu B_{\nu\rho} - \frac{1}{2} \mathcal{A}_\mu^i \zeta_{ij} \mathcal{F}_{\nu\rho}^j + (\text{cyclic perms.}) \right) \\
 & \left. \left(\partial^\mu B^{\nu\rho} - \frac{1}{2} \mathcal{A}^{i\mu} \zeta_{ij} \mathcal{F}^{j\nu\rho} + (\text{cyclic perms.}) \right) \right].
 \end{aligned}$$

Invariance to $O(E, E)$

What do we have here? A symmetry !

$$S' = S$$

- What did the R reduction showed us
- $O(E, E)$ symmetry
- What can we do next ?